Mat2002 Tutorial 3

Junwei Zha

January 26, 2026

Existence and Uniqueness theorem

Theorem (linear case)

Consider the linear differential equation:

$$\begin{cases} y' + p(t)y = q(t), \\ y(t_0) = y_0. \end{cases}$$

If $p(t)$ and $q(t)$ are continuous on an open interval I containing t_0, then there exists a unique solution to the initial value problem (IVP) on I.

Proof: The explicit solution is given by:

$$y = \frac{1}{\mu(t)} \left(\int \mu(t) q(t)\, dt + c \right),$$

where $\mu(t) = e^{\int p(t)\, dt}$ and c is determined by the initial condition.

Nonlinear Case

Theorem

Consider the nonlinear differential equation: $\begin{cases} y' = f(t, y), \\ y(t_0) = y_0. \end{cases}$

If f *and* $\frac{\partial f}{\partial y}$ *are continuous in a rectangle* $R = \{a < t < b, c < y < d\}$ *containing* (t_0, y_0)*, then there exists a unique solution to the IVP in some interval* $I = (t_0 - h, t_0 + h) \subseteq (a, b)$ *containing* t_0*.*

Remark

1. *For non-linear equations, the interval of existence* I *may be small than* (a, b)*, in this case, we call the solution "local". For linear equation, the solution is "global"*
2. *The continuity of* f_y *can be replace by a weaker condition, Lipschitz condition.*
3. *If only assume the continuity of* f*, the solution still exist, but may not unique.*

Example for Linear Case

Find the interval of existence of the solution without solving it.

1. $$\begin{cases} y' + t^2 y = \cos t, \\ y(0) = 1. \end{cases}$$

   Here, $p(t) = t^2$ and $q(t) = \cos t$ are continuous on $\mathbb{R}$, so the solution exists on $I = \mathbb{R}$.

2. $$\begin{cases} ty' + 2y = 2t^2, \\ y(-1) = 0. \end{cases}$$

   Rewrite as $y' + \frac{2}{t} y = 2t$, $t \neq 0$. Here, $p(t) = \frac{2}{t}$ and $q(t) = 2t$ are continuous on $(-\infty, 0) \cup (0, +\infty)$. The initial condition is at $t_0 = -1$, so the interval of existence is $(-\infty, 0)$

Example for Nonlinear Case

Example

This example can be viewed as an application of existence and uniqueness theorem:

$$\text{Solve}: \begin{cases} y' = sin(e^t y) \\ y(1) = 0. \end{cases}$$

Sol: It is impossible to solve an explicit solution for $y' = sin(e^t y)$.

However, we can observe that $y = 0$ is a constant solution for IVP.

On the other hand, note that $f = sin(e^t y)$ and $f_y = sin(e^t y)e^t$ are continuous on $\mathbb{R}^2$.

By existence and uniqueness theorem, the solution should be $y = 0$

Blow-up of Solution

The solution may only be "local"

Example

Solve $\begin{cases} y' = y^2, \\ y(0) = 1. \end{cases}$

Sol: Consider $y' = y^2$ this is separable, we can find $y = 0$ or $y = \frac{1}{1-t}$

Note that $f = y^2$ and $f_y = 2y$ are both continuous on $\mathbb{R}^2$ but the solution is "local", only exist on $(-\infty, 1)$ instead of $\mathbb{R}^2$, because y goes to $+\infty$ as $t \to 1^-$.

The solution blow up at $t = 1$

Loss of Uniqueness

Consider the IVP:

$$\begin{cases} y' = y^{\frac{2}{3}}, \\ y(0) = 0. \end{cases}$$

Sol:

- $y = 0$ is a trivial solution.
- Another solution is $y = \left(\frac{1}{3}t\right)^3$.
- In fact, for any $a > 0$, the function

$$y = \begin{cases} 0, & t < a, \\ \left(\frac{1}{3}(t-a)\right)^3, & t \geq a, \end{cases}$$

  is also a solution.

Remark: The uniqueness condition is not satisfied because $f_y = \frac{1}{5}y^{-\frac{1}{5}}$ is not continuous at $y = 0$.

Exercise

Find the interval of existence without solving the IVP:

$$\begin{cases} (t-3)y' + ln(t)y = t, \\ y(1) = 1 \end{cases}$$

Exercise

Find the interval of existence without solving the IVP:

$$\begin{cases} (t-3)y' + ln(t)y = t, \\ y(1) = 1 \end{cases}$$

Solution

$p(t) = \frac{lnt}{t-3}$, $q(t) = \frac{t}{t-3}$ *are continuous on* $(0,3) \cup (3,\infty)$.

The initial condition is at $t_0 = 1$ *hence the interval of existence is* $I = (0,3)$

Excercise

State where in the ty-plane the assumptions of Existence and Uniqueness theorem are satisfied for

$$\frac{dy}{dt} = \frac{1+t^2}{3y-y^2}$$

Excercise

$$\frac{dy}{dt} = \frac{1+t^2}{3y-y^2}$$

Solution

$$f(t,y) = \frac{1+t^2}{3y-y^2} = \frac{1+t^2}{y(3-y)}$$

and $$\frac{\partial f}{\partial y} = \frac{(0)\left(3y-y^2\right) - \left(1+t^2\right)(3-2y)}{\left(3y-y^2\right)^2} = -\frac{\left(1+t^2\right)(3-2y)}{y^2(3-y)^2}$$

Both f and $\partial f/\partial y$ are discontinuous at $y = 0$ and $y = 3$. Therefore, the assumptions of Uniqueness theorem are satisfied if $y \neq 0$ and $y \neq 3$.

Excercise

Solve the given initial value problem and determine how the interval in which the solution exists depends on the initial value y_0.

$$y' + y^3 = 0, y(0) = y_0$$

Excercise

Solution

For: $y' + y^3 = 0, y(0) = y_0$

Separate variables: $\frac{dy}{y^3} = -dt$

Integrate both sides: $-\frac{1}{2y^2} = -t + C_2$

Apply the initial condition now to determine C_2*:* $-\frac{1}{2y_0^2} = C_2$

Therefore, $y(t) = \frac{y_0}{\sqrt{2ty_0^2+1}}$

The positive root was chosen so that the initial condition remains satisfied.

$f(t, y) = -y^3$ *and* $\frac{\partial f}{\partial y} = -3y^2$

Both f *and* $\partial f/\partial y$ *are continuous everywhere except where the solution for* y *blows up. Since the initial condition is given at* $t = 0$*, a unique solution exists in an interval within*

$$-\frac{1}{2y_0^2} < t < \infty$$

Excercise

Compute the first 3 Picard iterates for the following initial value problem

$$\frac{dy}{dt} = 1 - y^3, \quad y(0) = 0$$

Excercise

$$\frac{dy}{dt} = 1 - y^3, \quad y(0) = 0$$

Solution

Note that the first 3 Picard iterates are $y_0(t), y_1(t), y_2(t)$, *and recall the iteration formula given by*

$$y_{n+1}(t) = y_0 + \int_{y_0}^{t} f\left(s, y_n(s)\right) ds$$

where in this question $y_0 = 0$ *and* $t_0 = 0$. *Then the formula reduces to*
$y_{n+1}(t) = \int_0^t f(s, y(s)) ds = \int_0^t \left(1 - y_n^3(s)\right) ds$
Thus, we can compute $y_0(t) = 0$,
$y_1(t) = \int_0^t \left(1 - y_0^3(s)\right) ds = \int_0^t 1 ds = t$
$y_2(t) = \int_0^t \left(1 - y_1^3(s)\right) ds = \int_0^t \left(1 - t^3\right) ds = t - \frac{t^4}{4}$